A knowledge graph approach for interpretable prediction of pathogenic genetic interactions

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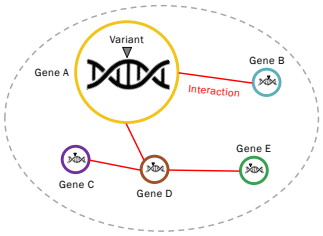
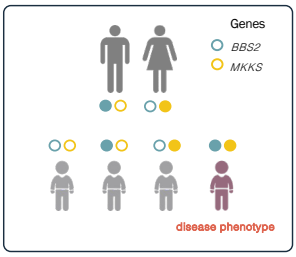
Abstract

- **Complex genetic diseases** are often caused by a **combination of variants and genes**. The current methods predicting such pathogenic variant/gene combinations can show good performance, but provide limited interpretability, hindering clinicians and researchers trust.
- This work provides a new resource to contextualize known pathogenic gene interactions in the form of a **knowledge graph** and a predictive method that can leverage knowledge graph patterns as rules to generate **interpretable** predictions of pathogenic gene pairs.

Gene interactions and oligogenic diseases

Oligogenic diseases are caused by the interaction between variants located in a few different genes. This interaction can be the direct cause of a disease phenotype or modulate the type and severity of disease symptoms.

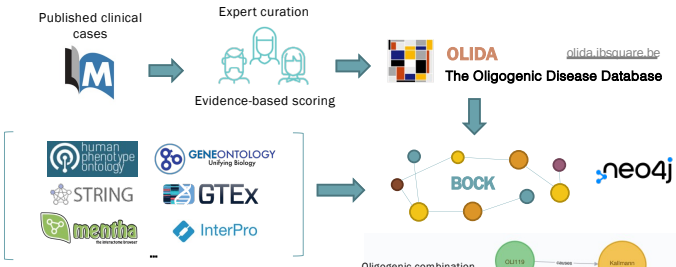




In **digenic inheritance**, the variants at **two loci** explain the phenotype of some patients and their relatives more clearly than the variant(s) at one locus alone (*Schäffer, 2013*).

On this pedigree, two variants in genes **BBS2** and **MKKS** explain more clearly the **Bardet-Biedl syndrome** of the affected child.

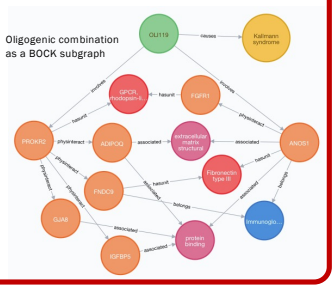
A knowledge graph of oligogenic combinations



Information about oligogenic diseases has been **expertly curated** from the medical literature and gathered in the **OLIDA database** (*Nachtegael, Gravel et al. 2022*).

A **knowledge graph**, BOCK, was created by integrating OLIDA with public biological networks and ontology resources. It contains **~83k nodes** (10 types) and **~2.6m edges** (17 types).

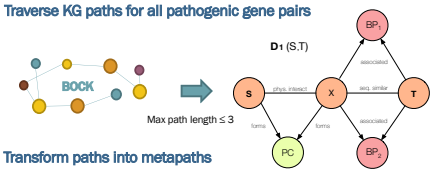
Oligogenic combination as a BOCK subgraph



Interpretable classification of pathogenic gene pairs based on metapath association rules

Mining patterns in the knowledge graph

Traverse KG paths for all pathogenic gene pairs



Transform paths into metapaths

M1: Gene — phys.interact — Gene — seq.similar — Gene
M2: Gene — forms — ProteinComplex — forms — Gene — seq.similar — Gene
M3: Gene — phys.interact — Gene — associated — BiologicalProcess — associated — Gene

Weighted frequent pattern mining with node-based unifications

Weighted instance table

D	w	Metapaths
D_1	1.0	M1, M2, M3
D_2	1.0	M2, M4, M7, M8
D_3	0.6	M2, M3, M8
D_4	0.6	M2, M3, M8, M9

Frequent metapath patterns

R	Pattern	Support
R_1	M2 & M3	2.2
R_2	M2 & M8	2.2
R_3	M3 & M8	1.2
R_4	M2 & M3 & M8	1.2

Add unification on common nodes

	R1 common nodes
D_1	X (Gene)
D_3	Y (Gene)

Crossover nodes between M2 & M3

R_5	M2 & M3 & M2.G := M3.G	1.6

Pathogenic gene pairs (441 pairs) are selected in the knowledge graph and all paths of length ≤ 3 are traversed and transformed into **metapaths**. A **weighted apriori algorithm** discovers **rules** based on the frequent occurrence of metapaths in gene pairs. **Unifications** are mined based on common node occurrence between metapaths.

Learning an interpretable model

Reducing noise with path confidence thresholding

	M2	M3
D_1	[0.6]	[0.8, 0.3]
D_3	[0.5, 0.8]	[0.7]

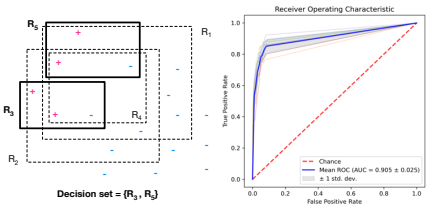
Maximize support — Minimize #paths

R_5	M2 ≥ 0.6 & M3 ≥ 0.7 & M2.G := M3.G	1.6

Path confidence score:
- Edge scores
- Node typed-degree centrality / Term IC

Finding optimal thresholds with a **differential evolution algorithm.**

Learning a decision set identifying pathogenic gene pairs



Rules are refined with **path confidence score thresholds** optimized to select only **high-quality paths** while maximizing the rule support. A **decision set model** is trained based on the pre-mined rules with a **weighted set cover algorithm**. Negative examples are selected based on gene pairs frequently mutated in healthy controls (1KGP).

Prediction graph explanations

From a knowledge graph hairball...

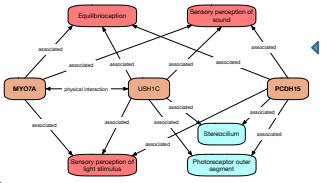
Subgraph from all paths of length ≤ 3 in BOCK for:
Gene pair: MYO7A-CDH15
Causing: Usher Syndrome I

(183 paths ; 80 nodes)

Matching rule (pl(+) = 0.95)

(Gene)-associated-(BP_1)-associated-(Gene) ≥ 0
& (Gene)-phys.interact-(Gene)-associated-(BP_2)-associated-(Gene) ≥ 0.3
& (Gene)-phys.interact-(Gene)-associated-(CC)-associated-(Gene) ≥ 0.4
& (BP_1) = (BP_2)

... to a graphical explanation



BOCK query result of the matching rule for MYO7A-PCDH15.

(8 paths ; 8 nodes)

The decision set model is **fully interpretable**. Each positive prediction comes with their matched rule(s). The rules can then be transformed into a **graphical explanation** by querying the knowledge graph. The resulting subgraph contains a subset of paths highlighting potential mediator genes and underlying biological mechanisms.

Conclusions

- This work demonstrates the potential of **knowledge graph-based rules** to identify pathogenic gene pairs, by exploiting the direct and indirect relationships between genes at different biological levels.
- Interpretable machine learning based on knowledge graph is a promising approach to provide **explanations users will understand and trust**, by enabling the context-aware assessment of predictions (fact checking) and the generation of new hypotheses.